

Individual Pathways

Activity 7.1

Revising chemical reactions

- Heat is taken in when ammonium nitrate is dissolved in water. This is an example of
 - neutralisation.
 - combustion.
 - exothermic change.
 - endothermic change.
- Which of these results in **only** a physical change?
 - Leaving bread to go mouldy
 - Adding bicarbonate of soda to vinegar
 - Mixing salt with water
 - Adding magnesium to hydrochloric acid
- When an acid and base react together, which substances would you always expect to be produced?
 - A salt and oxygen gas
 - A salt and hydrogen gas
 - A salt and water
 - Water, a salt and carbon dioxide
- Acids generally
 - taste bitter.
 - taste sour.
 - feel slippery.
 - are coloured.

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5. The complete combustion of the fossil fuel methane produces

- (a) carbon dioxide and hydrogen gas.
- (b) carbon dioxide and water.
- (c) a salt and hydrogen gas.
- (d) a salt and carbon dioxide gas.

6. Use the listed words to complete the sentences below.

absorb	alkalis	antacid	bitter	broken	colour
combustion	conservation	constant	electrons	energy	greater
lower	oxidation	oxygen	products	sour	water

- (a) A chemical reaction occurs when bonds are and the atoms form bonds in new combinations to form the products.
- (b) The law of mass is obeyed in a chemical reaction. The mass of the reactants equals the mass of the
- (c) Lavoisier showed that a compound always contains the same relative amounts of each element no matter how the compound was formed. This is the law of proportions.
- (d) Exothermic reactions release heat. Endothermic reactions heat. is an example of an exothermic reaction.
- (e) Acids are corrosive substances. Acids have a taste and attack reactive metals.
- (f) Bases have a taste and dissolve fatty materials. Bases that dissolve in water are called
- (g) The pH of a solution is a measure of its acidity or basicity. The pH of neutral water is 7. Acidic solutions have a pH than 7 and basic solutions have a pH than 7.
- (h) Acid-base indicators change as the pH of the solution changes.
- (i) Acids neutralise bases to form salts and The excess hydrochloric acid in your stomach can be neutralised by taking an
- (j) Combustion is a chemical reaction in which a substance combines with and produces such as heat and light.
- (k) Combustion reactions are examples of reactions in which are transferred from a reactant.

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7. In chemical reactions, mass is neither gained nor lost. The total mass of all the reactants equals the total mass of all the products. Atoms are just rearranged into different substances. Use this information to complete the following questions.

(a) Potassium chlorate $\rightarrow$ potassium chloride + oxygen

If 500 g of potassium chlorate decomposes and produces 303 g of potassium chloride, how many grams of oxygen are produced?

.....

(b) Iron + oxygen $\rightarrow$ iron oxide

How many grams of oxygen are needed to react with 350 g of iron to produce 500 g of iron oxide?

.....

8. The three common bases used in antacid preparations are aluminium hydroxide, magnesium hydroxide and magnesium carbonate. When eaten, an antacid helps to neutralise excess hydrochloric acid in the stomach. Aluminium chloride or magnesium chloride are the salts produced. Write word equations for each of these neutralisation reactions.

(a) aluminium hydroxide +

(b) magnesium hydroxide +

(c) magnesium carbonate +

9. Here are some results of solutions tested with universal indicator paper:

Solution	Colour
Sulfuric acid	Red
Metal polish	Dark blue
Washing up liquid	Yellow
Milk of magnesia	Light blue
Oven cleaner	Purple
Car battery acid	Pink

Arrange the solutions in order of their pH, starting with the one with the **lowest** pH.

.....
.....

Answers

1. d
2. c
3. c
4. b
5. b

6.
 - (a) A chemical reaction occurs when bonds are **broken** and the atoms form bonds in new combinations to form the products.
 - (b) The law of mass **conservation** is obeyed in a chemical reaction. The mass of the reactants equals the mass of the **products**.
 - (c) Lavoisier showed that a compound always contains the same relative amounts of each element no matter how the compound was formed. This is the law of **constant** proportions.
 - (d) Exothermic reactions release heat. Endothermic reactions **absorb** heat. **Combustion** is an example of an exothermic reaction.
 - (e) Acids are corrosive substances. Acids have a **sour** taste and attack reactive metals.
 - (f) Bases have a **bitter** taste and dissolve fatty materials. Bases that dissolve in water are called **alkalis**.
 - (g) The pH of a solution is a measure of its acidity or basicity. The pH of neutral water is 7. Acidic solutions have a pH **lower** than 7 and basic solutions have a pH **greater** than 7.
 - (h) Acid-base indicators change **colour** as the pH of the solution changes.
 - (i) Acids neutralise bases to form salts and **water**. The excess hydrochloric acid in your stomach can be neutralised by taking an **antacid**.
 - (j) Combustion is a chemical reaction in which a substance combines with **oxygen** and produces **energy** such as heat and light.
 - (k) Combustion reactions are examples of **oxidation** reactions in which **electrons** are transferred from a reactant.

7.
 - (a) Mass of oxygen = $500 - 303 = 197$ g
 - (b) Mass of oxygen = $500 - 350 = 150$ g

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8.

- (a) aluminium hydroxide + **hydrochloric acid** → **aluminium chloride** + **water**
- (b) magnesium hydroxide + **hydrochloric acid** → **magnesium chloride** + **water**
- (c) magnesium carbonate + **hydrochloric acid** → **magnesium chloride** + **water** + **carbon dioxide**

9. Sulfuric acid, car battery acid, washing up liquid, milk of magnesia, metal polish, oven cleaner